Total Knee Arthroplasty *Prodigy Anesthesia*

A total knee arthroplasty (TKA) is performed to replace the knee joint surfaces with metal and plastic components. The surgery can also be performed on a single compartment of the knee as a medial or lateral unicompartmental knee arthroplasty.

During a TKA, a longitudinal incision is made along the midline of the knee and the femur, patella, and tibia are excised with bone saws.

It is more important to note that the incidence of deep vein thrombosis without prophylactic treatment is between 50-75% and drops to 2-3% if low-molecular weight heparin, coumadin, sequential compression devices, anti-embolism stockings, or some combination of the above is utilized.





The patient is positioned supine and a thigh tourniquet is typically used. It is common for the blood pressure to decrease by 20% or so after tourniquet deflation. The tourniquet should be inflated to about 100 mmHg above the systolic blood pressure. Tourniquet times should not exceed 2 hours unless a 30 minute reperfusion interval is allowed before re-inflating the tourniquet. Deflation of the tourniquet can be accompanied by a drop in central venous pressure and systemic blood pressure. The heart rate typically increases and the core body temperature will fall. As accumulated metabolic wastes return to the systemic circulation there is an increase in ETCO2, PaCO2, serum potassium, and serum lactate levels.

The most common diagnosis for a patient presenting for total knee arthroplasty is osteoarthritis. If rheumatoid arthritis is the cause, the patient will need evaluation for involvement of the cervical spine prior to the induction of anesthesia as well as an exploration of the common sequelae of rheumatoid arthritis such as pericardial effusions, cardiac valve fibrosis, and conduction abnormalities.

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Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 1038-1052.

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